Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

TECHNICAL CATALOGUE · HAND-DRIVEN · BAKELITE BODY

CIE/555

Compact hand-driven insulation tester in moulded bakelite

Direct-reading insulation resistance testing

A compact, portable instrument for direct reading of insulation resistance on electrical operating machines, rotating plant equipment, stationary apparatus, insulation control and switching equipment, distribution networks, communication cables, insulators, transformers, industrial installations, cables, electrical components, wiring and other devices capable of carrying the rated voltage. It is a versatile instrument for locating intermittent shorts, defective electrical connections, insulation failure and conductor failure, and for preventing costly repairs and breakdowns.



CIE/555 · ILLUSTRATIVE IMAGE

11	4	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

100 V · 250 V · 500 V · 1000 V

01 HIGHLIGHTS

- ✓ Moulded high-grade bakelite body
- ✓ Collapsible driven handle, folds flush into the casing when not in use
- ✓ High-sensitivity cross-coil indicator
- ✓ Two push buttons for connection, accepting standard cable or 4 mm plug
- ✓ Accurate and quick response: 3 seconds
- ✓ Silent generator
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Rubber shoes hold the surface and give stability
- ✓ Graceful appearance
- ✓ Conforms to IS 2992:1987

02 TYPICAL APPLICATIONS

- Insulation testing of electrical operating machines
- Motors, generators, and other rotating machinery
- Insulation control and switching equipment testing
- Distribution network testing
- Communication cable insulation testing
- Insulator and transformer insulation checks
- Industrial installation and wiring checks
- Locating intermittent shorts and defective electrical connections

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03 PRINCIPLE AND CONSTRUCTION

PRINCIPLE

A hand-driven clutch-type generator and a high-sensitivity cross-coil indicator. The generated A.C. voltage is converted to D.C. voltage by silicon diodes.

CONSTRUCTION

Moulded in high-grade bakelite. The instrument is dust-free: rubber shoes hold the surface and give it stability.

04 SALIENT FEATURES

Collapsible handle

The driven handle folds flush into the casing when the instrument is not in operation.

Chuck nut locking

A chuck nut locking arrangement prevents anti-clockwise rotation of the handle.

Push-button connection

Two push buttons are provided for connection, designed to accommodate a standard cable or a 4 mm plug.

Quick response

The response time is 3 seconds.

05 PERFORMANCE AND TEST SPECIFICATION

Standard	IS 2992:1987
Indicator	High-sensitivity cross-coil
Scale length	75 mm (approx.)
Accuracy	Conforms to IS 2992:1987
Generator speed	160 r.p.m.
Response time	3 seconds
High voltage test	Withstands minimum 2000 V or (rated voltage + 1000) V A.C. (rms) between the electrical circuit and the handle for one minute
Insulation resistance of the instrument	Better than 50 MΩ at 500 V between the electrical circuit and the containing case

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06 CONSTRUCTION AND PHYSICAL SPECIFICATION

Body	Moulded high-grade bakelite
Weight	1.4 kg (approx.)
Dimensions	155 × 104 × 70 mm (approx.)

07 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
100 V	0–10, 0–20
250 V	0–20, 0–50
500 V	0–50, 0–100, 0–200, 0–500
1000 V	0–100, 0–200, 0–500

Ranges shown are the standard builds. Any other range can be manufactured against specific requirements.

08 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Carrying case (bakelite body)
- Testing leads, 1 m

09 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/555, 500 V / 0–500 MΩ**. Any range not listed can be manufactured against specific requirements.

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10 CIE HAND-DRIVEN AND MOTOR-OPERATED INSULATION TESTERS COMPARED

	CIE/444	CIE/666	CIE/555	CIE/777	CIE/777 HM
Drive	Hand-driven	Hand-driven	**Hand-driven**	Mains motor, 230 V, 50 Hz	Mains motor, 230 V, 50 Hz, or hand cranking
Body	Anti-magnetic cast aluminium	Seasoned wood, sunmica finish	**Moulded bakelite**	Seasoned wooden cabinet, sunmica finish	Seasoned wooden cabinet, sunmica finish
Rated voltages (V)	100, 250, 500, 1000, 2500, 5000	100, 250, 500, 1000, 2500, 5000, 10,000	**100, 250, 500, 1000**	500, 1000, 2500, 5000, 10,000	500, 1000, 2500, 5000, 10,000
Standard range combinations	27	32	**11**	14	14
Multirange series	MR-1 to MR-4	MR-1 to MR-8	**Single range only**	MR-A to MR-E	MR-A to MR-E
Scale length	90 mm	100 mm	**75 mm**	100 mm	100 mm
Response time	3 seconds	3 seconds	**3 seconds**	3 seconds	3 seconds
Weight (approx.)	2.5 kg	4.1 kg	**1.4 kg**	9 kg (incl. top cover)	12 kg (incl. top cover)
Conforms to	IS 2992:1987	IS 2992:1987	**IS 2992:1987**	IS 2992:1987	IS 2992:1987

Rated voltages, standard ranges and multirange series as listed in this catalogue series. Weights and scale lengths are approximate.

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